

# Competitive Exam Practice 8: I/O Techniques

## GATE (Computer Science) Pattern

Auqib Hamid Lone

*Department of Computer Science & Engineering, GCET Ganderbal*

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type; TF = True/False.

**Q1** [*GATE CS 1987 Q1-viii, verified — GATEOverflow V3 p.65*] [**MCQ, 1 mark**] On receiving an interrupt from an I/O device, the CPU:

- (A) Halts for a predetermined time.
- (B) Hands over control of the address bus and data bus to the interrupting device.
- (C) Branches off to the interrupt service routine immediately.
- (D) Branches off to the interrupt service routine after completion of the current instruction.

**Q2** [*GATE CS 1995 Q1.3, verified — GATEOverflow V3 p.65*] [**MCQ, 1 mark**] In a vectored interrupt:

- (A) The branch address is assigned to a fixed location in memory.
- (B) The interrupting source supplies the branch information to the processor through an interrupt vector.
- (C) The branch address is obtained from a register in the processor.
- (D) None of the above.

**Q3** [*GATE CS 1998 Q1.20, verified — GATEOverflow V3 p.65*] [**MCQ, 1 mark**] Which of the following is true?

- (A) Unless enabled, a CPU will not be able to process interrupts.
- (B) Loop instructions cannot be interrupted till they complete.
- (C) A processor checks for interrupts before executing a new instruction.
- (D) Only level-triggered interrupts are possible on microprocessors.

**Q4** [*GATE CS 1996 Q1.24, verified — GATEOverflow V3 p.59*] [**MCQ, 1 mark**] For the daisy-chain scheme of connecting I/O devices, which of the following statements is true?

- (A) It gives non-uniform priority to various devices.
- (B) It gives uniform priority to all devices.
- (C) It is only useful for connecting slow devices to a processor device.

(D) It requires a separate interrupt pin on the processor for each device.

**Q5** [*GATE CS 1987 Q2a, verified — GATEOverflow V3 p.59*] [**TF, 1 mark**] State whether the following statement is TRUE or FALSE:

In a microprocessor-based system, if a bus (DMA) request and an interrupt request arrive simultaneously, the microprocessor attends first to the bus request.

**Q6** [*GATE CS 1987 Q2b, verified — GATEOverflow V3 p.59*] [**TF, 1 mark**] State whether the following statement is TRUE or FALSE:

Data transfer between a microprocessor and an I/O device is usually faster in a memory-mapped-I/O scheme than in an I/O-mapped-I/O scheme.

**Q7** [*GATE CS 1990 Q4-ii, verified — GATEOverflow V3 p.59*] [**TF, 2 marks**] State whether the following statement is TRUE or FALSE, with reason:

The data transfer between memory and I/O devices using programmed I/O is faster than interrupt-driven I/O.

**Q8** [*Original, GATE-pattern*] [**NAT, 2 marks**] A 2 GHz CPU reads input from a keyboard producing 10 keystrokes per second using a `READWAIT` polling loop of 2 instructions per failed iteration. The number of polling iterations the CPU executes *between* two consecutive keystrokes is \_\_\_\_\_.

**Q9** [*Original, GATE-pattern*] [**MCQ, 1 mark**] Despite being skew-immune and self-adapting to any device speed, most high-speed processor buses use synchronous rather than asynchronous protocols primarily because:

- (A) Synchronous buses require no shared bus clock.
- (B) Every asynchronous transfer pays two full round-trip delays, which a fast, similar-speed synchronous bus never pays.
- (C) Synchronous buses are immune to slow devices needing wait states.
- (D) Asynchronous protocols cannot support more than one bus master.

**Q10** [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a DMA transfer, *cycle stealing* refers to:

- (A) The CPU taking memory bus cycles away from the DMA controller.
- (B) The DMA controller taking memory bus cycles the CPU would otherwise have used, because both compete for the same bus.
- (C) The CPU executing one ISR invocation per stolen word.
- (D) The bus arbiter granting the bus to the CPU in preference to the DMA controller.

**Q11** [*Original, GATE-pattern*] [**MCQ, 2 marks**] An I/O processor (IOP) differs from a DMA controller in that it:

- (A) Moves the same volume of data with less extra hardware.
- (B) Executes a whole channel program of transfers, formats, and simple decisions independently, interrupting the CPU only when the entire program completes.
- (C) Never interrupts the CPU at all.
- (D) Does not need to become bus master via arbitration.